

Building and evaluation of a PBPK model for COMPOUND in healthy adults

Version	master-OSP12.2
based on <i>Model Snapshot and Evaluation Plan</i>	https://github.com/Open-Systems-Pharmacology/COMPOUND-Model/releases/tag/vmaster
OSP Version	12.2
Qualification Framework Version	3.5

This evaluation report and the corresponding PK-Sim project file are filed at:

<https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/>

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1 Introduction

Dabigatran is an oral direct thrombin inhibitor administered as a pharmacologically inactive prodrug, dabigatran etexilate. After absorption, it undergoes enzymatic conversion via carboxylesterases (CES1 in the liver and CES2 in the intestine) to the active moiety dabigatran. Dabigatran (etexilate) shows low oral bioavailability (~6–7%), moderate inter-individual variability, and is predominantly eliminated unchanged via renal excretion.

In addition to renal clearance, dabigatran undergoes glucuronidation, forming dabigatran acylglucuronide, which contributes to total active exposure. Absorption of dabigatran etexilate is strongly influenced by P-glycoprotein (P-gp), making transporter activity a key determinant of systemic exposure and drug–drug interaction potential.

The presented whole-body PBPK model is based on the model developed by Moj et al. (Moj 2019) and includes dabigatran, its prodrug dabigatran etexilate and the active metabolite dabigatran acylglucuronide. It was established to be used for DDI predictions as dabigatran etexilate is a widely used Pgp substrate in clinical DDI studies.

The herein presented model building and evaluation report evaluates the performance of the PBPK model for dabigatran in healthy adults.

The presented dabigatran PBPK model as well as the respective evaluation plan and evaluation report are provided open-source (<https://github.com/Open-Systems-Pharmacology/Dabigatran-Model>). The DDI performance is documented in the DDI qualification report (<https://github.com/Open-Systems-Pharmacology/COMPOUND-Model>).

2 Methods

2.1 Modeling Strategy

The general concept of building a PBPK model has previously been described by Kuepfer et al. ([Kuepfer 2016](#)). Regarding the relevant anthropometric (height, weight) and physiological parameters (e.g. blood flows, organ volumes, binding protein concentrations, hematocrit, cardiac output) in adults was gathered from the literature and has been previously published ([PK-Sim Ontogeny Database Version 7.3](#)). The information was incorporated into PK-Sim® and was used as default values for the simulations in adults.

The applied activity and variability of plasma proteins and active processes that are integrated into PK-Sim® are described in the publicly available PK-Sim® Ontogeny Database Version 7.3 ([Schlender 2016](#)) or otherwise referenced for the specific process.

The PBPK model was developed using a stepwise, mechanistic modeling strategy. First, an intravenous dabigatran model was established to describe distribution and elimination processes, including UGT2B15-mediated glucuronidation of dabigatran to dabigatran glucuronide as well as the renal clearance of both.

Second, the model was extended to include the orally administered prodrug dabigatran etexilate. This step required incorporation of Pgp-mediated transport affecting its absorption and CES1/CES2-mediated prodrug activation to produce dabigatran.

The respective contributions of CES1 and CES2 were informed by data from the RE-LY trial ([Paré 2013](#)), reporting that patients carrying 2 minor CES1 alleles (modeled as loss of CES1 activity) show a 28% reduction of their total dabigatran plasma concentration trough values (total dabigatran = dabigatran + dabigatran glucuronide).

While the original model considered data of 2 different clinical DDI studies (rifampicin-mediated Pgp induction ([Härtter 2012](#)) and clarithromycin-mediated Pgp inhibition ([Delavenne 2013](#))) for the model update data of 7 further clinical DDI trials were used, adding Pgp inhibition by rifampicin (n=2), itraconazole (n=1), verapamil (n=1), and additional clarithromycin studies (n=3) (please see the DDI qualification report (<https://github.com/Open-Systems-Pharmacology/COMPOUND-Model>)).

Unknown parameters were identified using the PK-Sim® Parameter Identification module. Model evaluation included comparison of predicted and observed concentration–time profiles, goodness-of-fit plots, and geometric mean fold error (GMFE) analysis. Model performance was considered acceptable if predictions were within a two-fold range of observed data.

Details about input data (physicochemical, *in vitro* and clinical) can be found in [Section 2.2](#).

Details about the structural model and its parameters can be found in [Section 2.3](#).

2.2 Data

2.2.1 In vitro / physico-chemical Data

A literature search was performed to collect available information on physicochemical properties of dapagliflozin. The obtained information from literature is summarized in the table below.

Parameter	Unit	Value	Source	Description
MW	g/mol		Dummy YEAR	Molecular weight
pK _a			DrugBank DB06292	Acid dissociation constant
Solubility (pH)			Dummy YEAR	Aqueous Solubility, FaSSIF, ...
logP			Dummy YEAR (experimental)	Partition coefficient between octanol and water
fu	%		Dummy YEAR	Fraction unbound in plasma
B/P ratio			Dummy YEAR	Blood to plasma ratio
...				

2.2.2 Clinical Data

A literature search was performed to collect available clinical data on dabigatran in healthy adults.

2.2.2.1 Model Building

The following studies were used for model building (training data):

Source	Arm / Treatment / Information used for model building
1160-0005 internal clinical trial report 2001	Intravenous infusion of 0.1 mg (single dose)
1160-0005 internal clinical trial report 2001	Intravenous infusion of 1.0 mg (single dose)
1160-0005 internal clinical trial report 2001	Intravenous infusion of 5.0 mg (single dose)
1160-0061 internal clinical trial report 2006	Oral capsule administration of 110 mg and 150 mg (twice daily)
1160-0194 internal clinical trial report 2014	Oral solution administration of 150 mg (twice daily)
1160-0194 internal clinical trial report 2014	Oral capsule administration of 150 mg (twice daily)
Blech 2008	Oral solution administration of 200 mg (single dose)
Härtter 2012	Oral capsule administration of 150 mg (single dose)

2.2.2.2 Model Verification

The following studies were used for model verification:

Source	Arm / Treatment / Information used for model building
1160-0074 internal clinical trial report 2009	Oral capsule administration of 150 mg (single dose)
1160-0082 internal clinical trial report 2008	Oral capsule administration of 150 mg (single dose)
Delavenne 2013	Oral capsule administration of 300 mg (single dose)

2.3 Model Parameters and Assumptions

2.3.1 Absorption

...

2.3.2 Distribution

...

After testing the available organ-plasma partition coefficient and cell permeability calculation methods built in PK-Sim, observed clinical data was best described by choosing the partition coefficient calculation by `Rodgers and Rowland` and cellular permeability calculation by `PK-Sim Standard`.

...

2.3.3 Metabolism and Elimination

...

2.3.4 Automated Parameter Identification

This is the result of the final parameter identification.

Model Parameter	Optimized Value	Unit
<code>PK-Sim parameter 1</code>		
<code>PK-Sim parameter 2</code>		
<code>PK-Sim parameter 3</code>		
<code>PK-Sim parameter 4</code>		

3 Results and Discussion

The PBPK model for COMPOUND was developed and verified with clinical pharmacokinetic data.

The model was evaluated covering data from studies including in particular

- ...
- ...

The model quantifies ...

The next sections show:

1. the final model parameters for the building blocks: [Section 3.1](#).
2. the overall goodness of fit: [Section 3.2](#).
3. simulated vs. observed concentration-time profiles for the clinical studies used for model building and for model verification: [Section 3.3](#).

3.1 Final input parameters

The compound parameter values of the final PBPK model are illustrated below.

Compound: DabiEtex

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	1.8 mg/ml		Measurement	True
Reference pH	7		Measurement	True
Lipophilicity	1.8088312812 Log Units	Parameter Identification-Parameter Identification-Value updated from '#11 - P-gp 0.46, DabiGluc logP -0.80, CESlink' on 2026-03-05 10:32	Measurement	True
Fraction unbound (plasma, reference value)	0.07		Measurement	True
Permeability	NaN cm/min		FIT	False
Specific intestinal permeability (transcellular)	1.85E-07 cm/min	Unknown	Fit	True
Is small molecule	Yes			
Molecular weight	627.73 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Transport Protein: ABCB1-FIT

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Transporter concentration	1 µmol/l	
Vmax	0 µmol/l/min	
Km	0.08 µmol/l	
kcat	0.46 1/min	Parameter Identification-Parameter Identification-Value updated from '#11 - P-gp 0.46, DabiGluc logP -0.80, CESlink' on 2026-03-05 10:32

Metabolizing Enzyme: CES1-FIT

Molecule: CES1

Metabolite: Dabigatran

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	24.9 µmol/l	
kcat	10.8149677921 1/min	Parameter Identification-Parameter Identification-Value updated from '#11 - P-gp 0.46, DabiGluc logP -0.80, CESlink' on 2026-03-05 10:32

Metabolizing Enzyme: CES2-FIT

Molecule: CES2

Metabolite: Dabigatran

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	5.5 µmol/l	
kcat	0.7209978528 1/min	Parameter Identification-Parameter Identification-Value updated from '#11 - P-gp 0.46, DabiGluc logP -0.80, CESlink' on 2026-03-05 10:32

Systemic Process: Glomerular Filtration-DabiEtex

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Compound: DabiGluc

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	1.8 mg/ml		Measurement	True
Reference pH	7		Measurement	True
Lipophilicity	-0.8 Log Units	Parameter Identification-Parameter Identification-Value updated from '#11 - P-gp 0.46, DabiGluc logP -0.80, CESlink' on 2026-03-04 18:22	Measurement	True
Fraction unbound (plasma, reference value)	0.65		Measurement	True
Permeability	NaN cm/min		FIT	False
Is small molecule	Yes			
Molecular weight	647.23 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-DabiGluc

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	6.36	

Compound: Dabigatran

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	17 mg/l		Measurement	True
Reference pH	7		Measurement	True
Lipophilicity	1.1436481967 Log Units	Parameter Identification-Parameter Identification-Value updated from '#7 - iv (logP, logP, UGT, GFR)_best iv' on 2025-08-30 15:56	Measurement	True
Fraction unbound (plasma, reference value)	0.65		Measurement	True
Permeability	4.6547688496E-06 dm/min	Parameter Identification-Parameter Identification-Value updated from '#1 update' on 2025-08-14 16:39	FIT	False
Is small molecule	Yes			
Molecular weight	471.51 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-Dabi

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Metabolizing Enzyme: UGT2B15-FIT

Molecule: UGT2B15

Metabolite: DabiGluc

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	512 μ mol/l	
kcat	4.2993761606 1/min	Parameter Identification-Parameter Identification-Value updated from '#7 - iv (logP, logP, UGT, GFR)_best iv' on 2025-08-30 15:56

Formulation: DabiEtex_Capsule

Type: Weibull

Parameters

Name	Value	Value Origin
Dissolution time (50% dissolved)	1.0579 min	
Lag time	0 h	
Dissolution shape	0.2628	
Use as suspension	No	

3.2 Diagnostic Plots

Below you find the goodness-of-fit visual diagnostic plots for the PBPK model performance of all data used presented in [Section 2.2.2](#).

The first plot shows observed versus simulated plasma concentration, the second weighted residuals versus time.

Table 3-1: GMFE for Goodness of fit plot for concentration in plasma

Group	GMFE
Intravenous infusion	1.16
Oral capsule	1.35
Oral capsule - Verification dataset	1.83
Oral solution	1.92
All	1.50

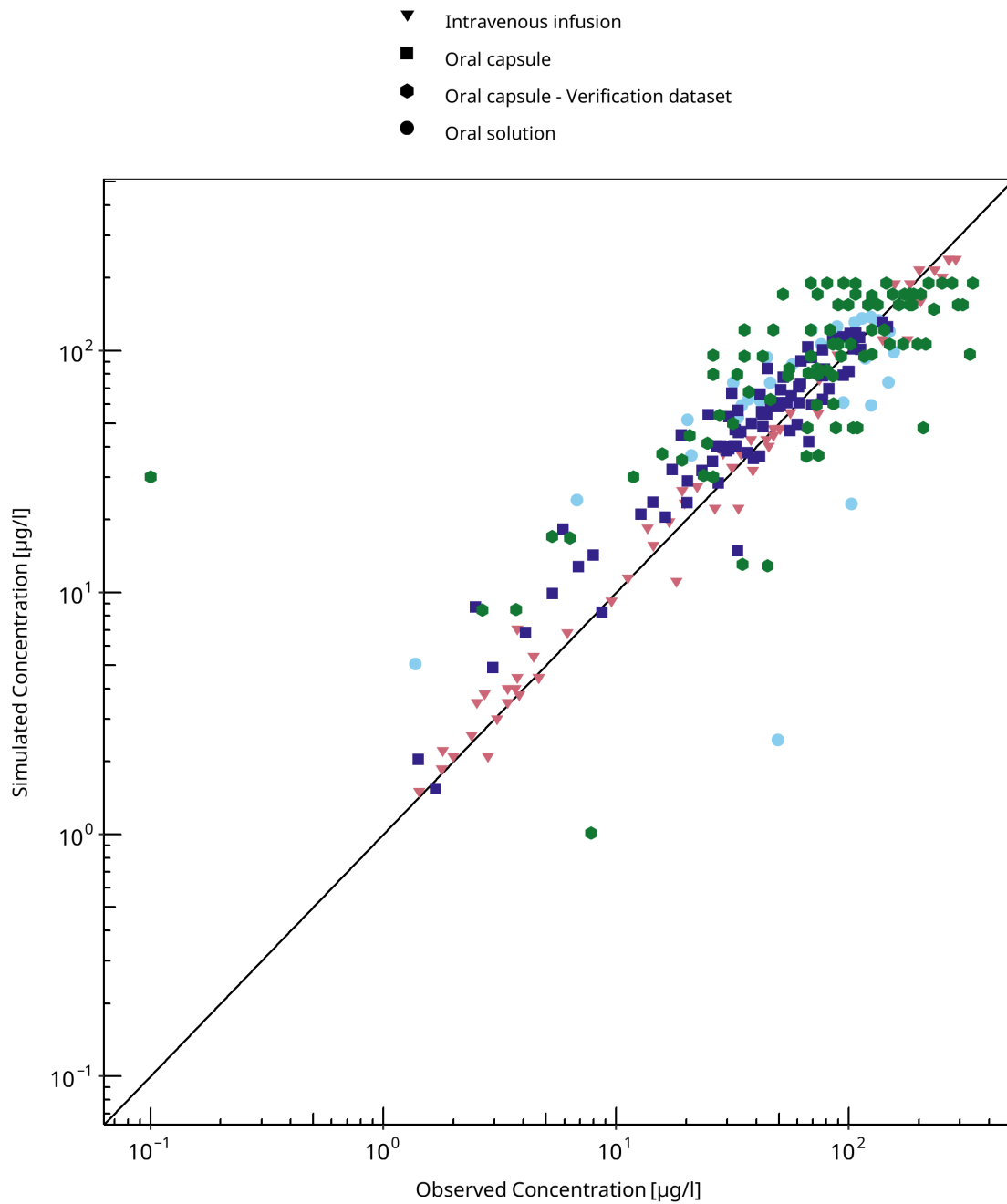


Figure 3-1: Goodness of fit plot for concentration in plasma

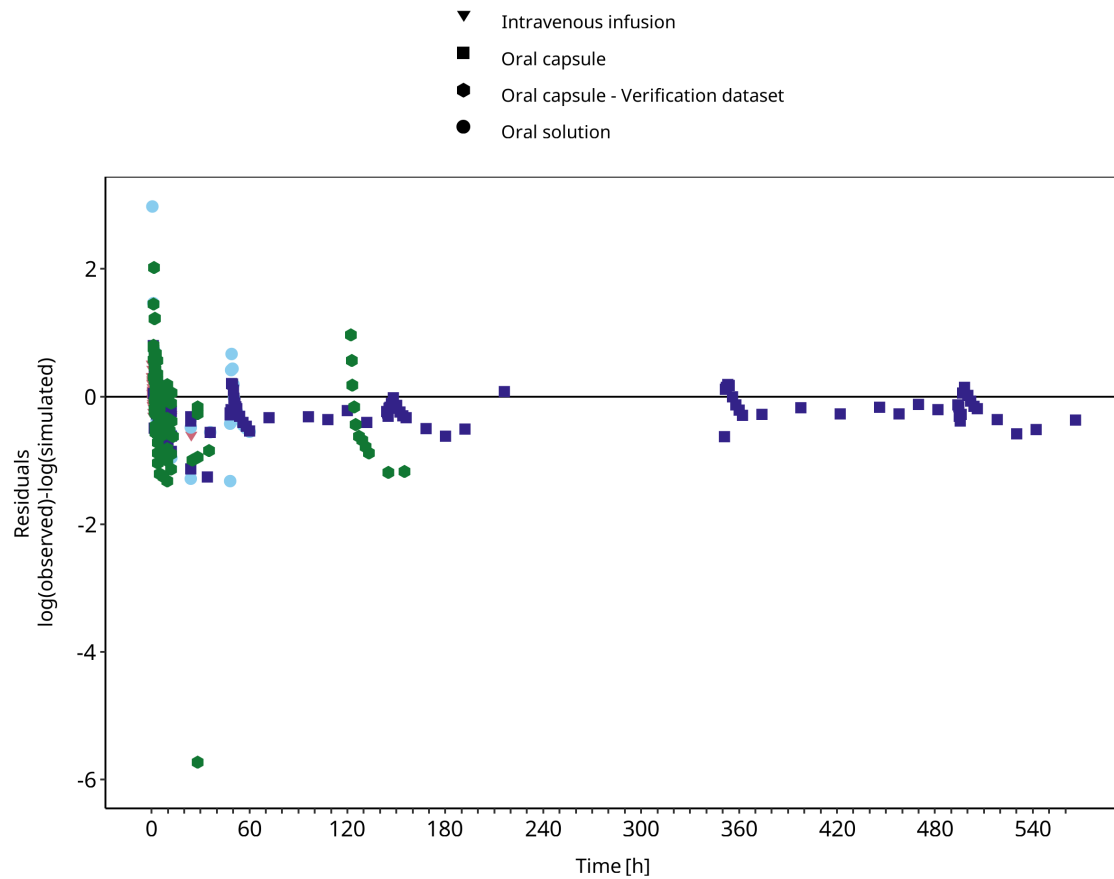


Figure 3-2: Goodness of fit plot for concentration in plasma

3.3 Concentration-Time Profiles

Simulated versus observed concentration-time profiles of all data listed in [Section 2.2.2](#) are presented below.

3.3.1 Model Building

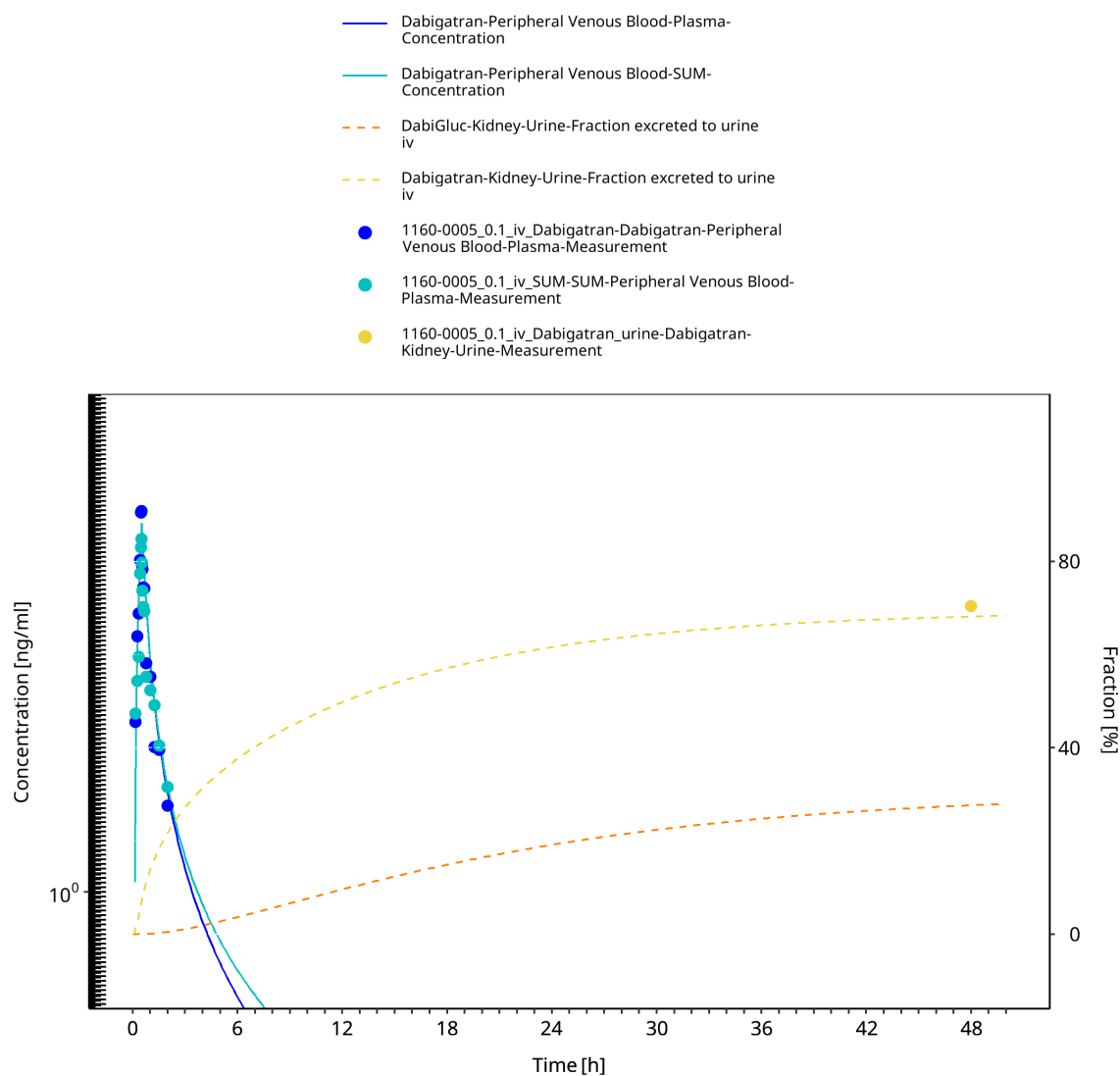


Figure 3-3: 1160-0005 - Dabigatran iv 0.1 mg - log

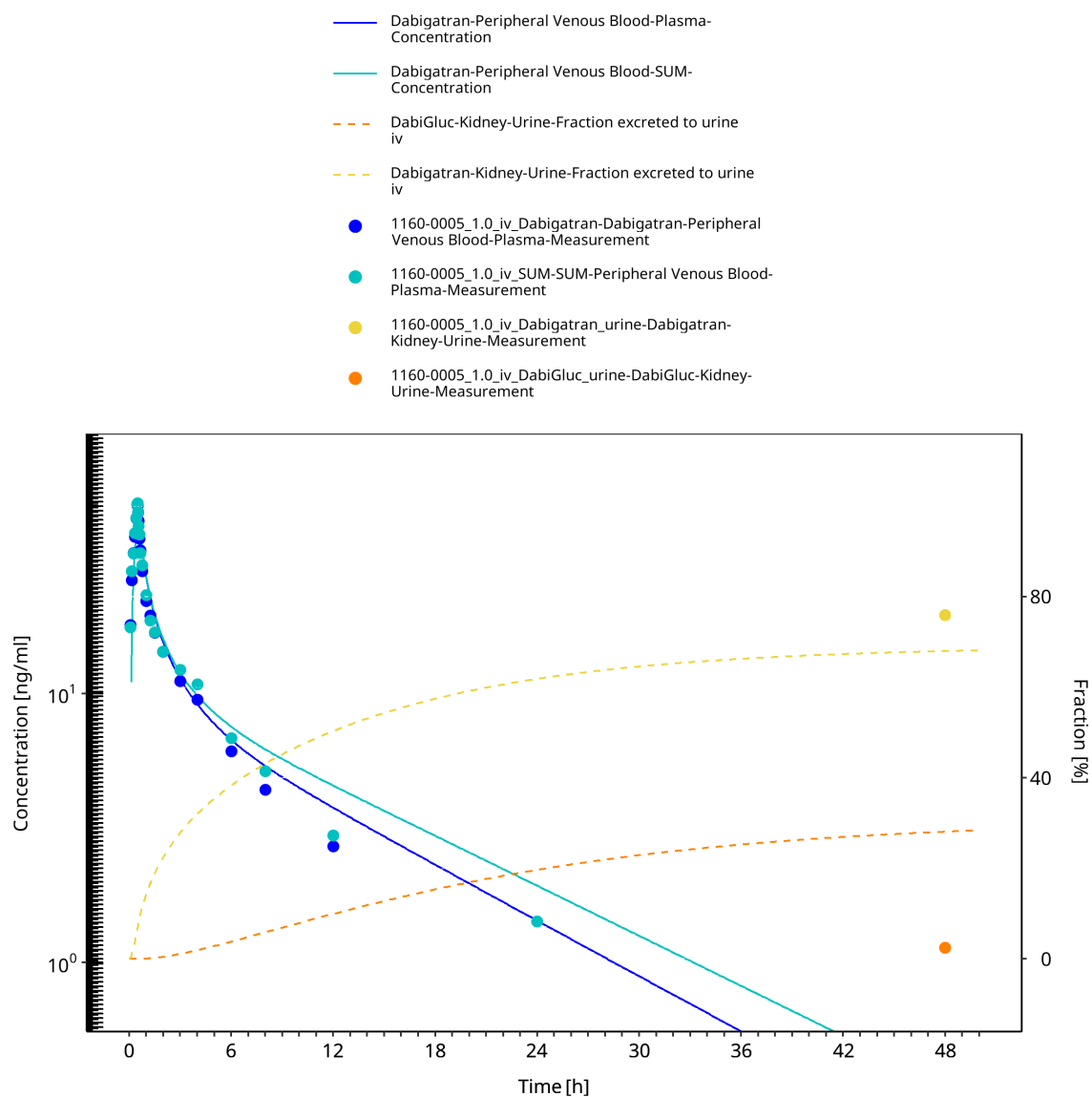


Figure 3-4: 1160-0005 - Dabigatran iv 1.0 mg - log

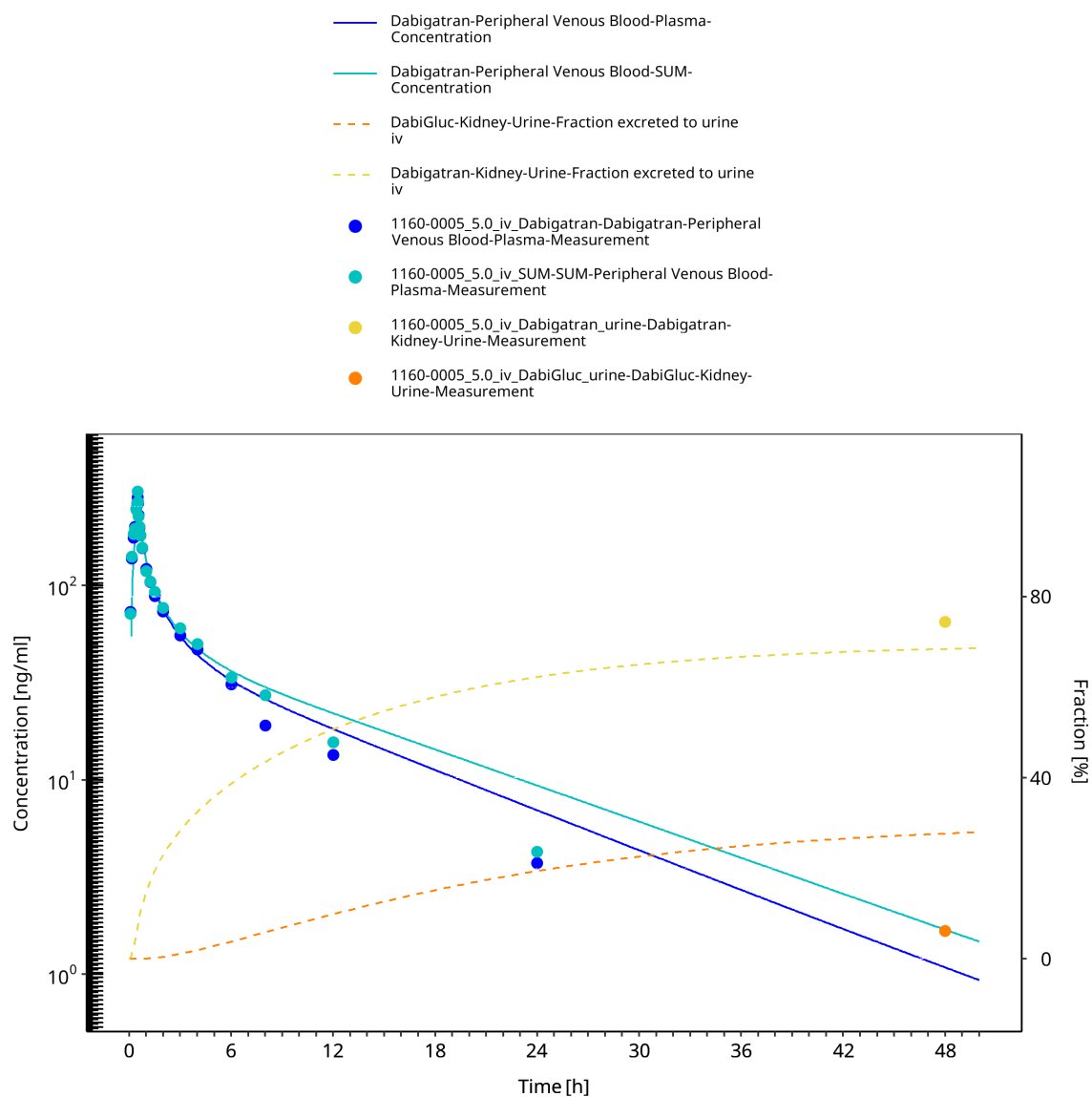


Figure 3-5: 1160-0005 - Dabigatran iv 5.0 mg - log

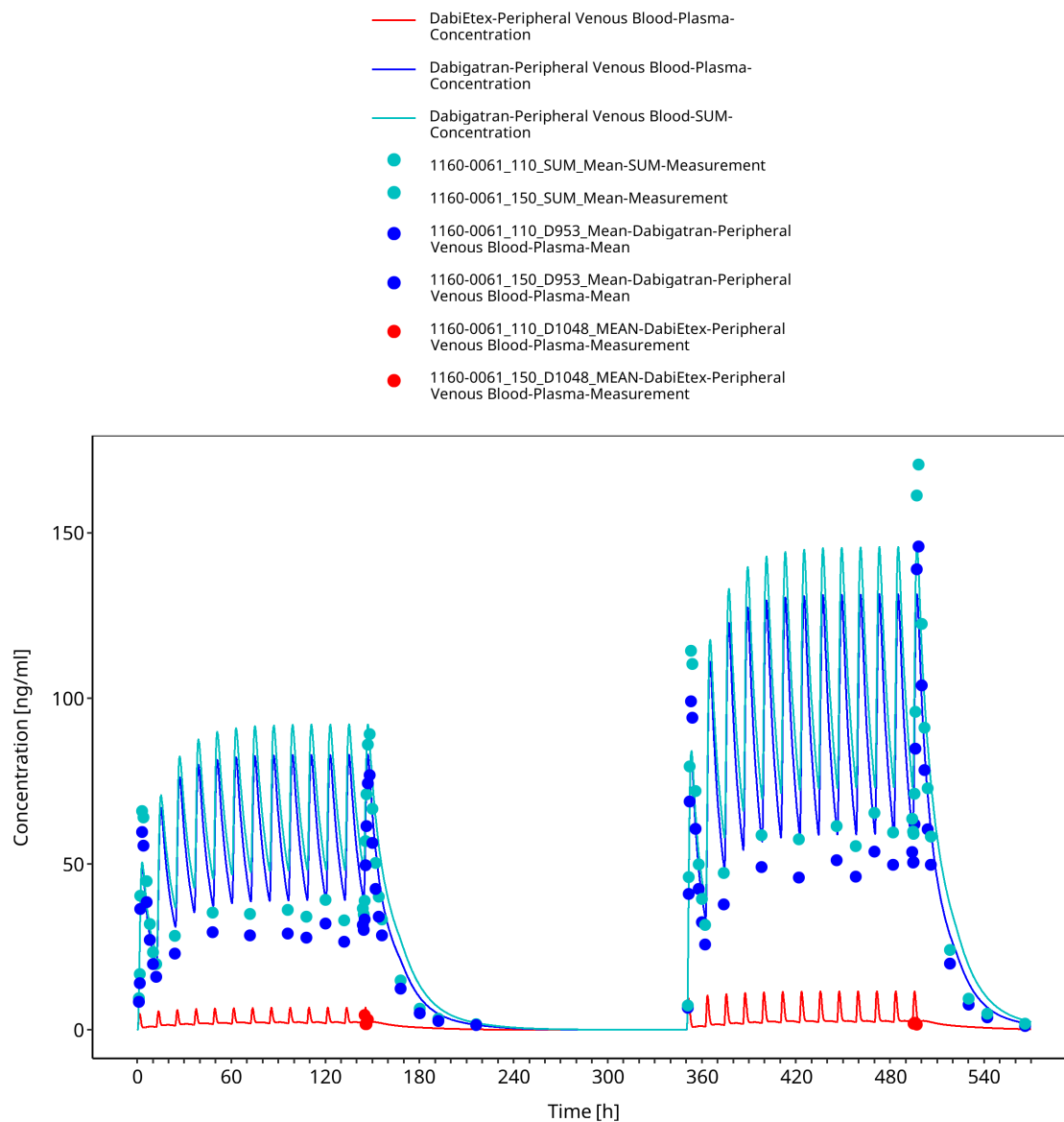


Figure 3-6: 1160-0061 - Dabigatran 110+150 mg BID - lin

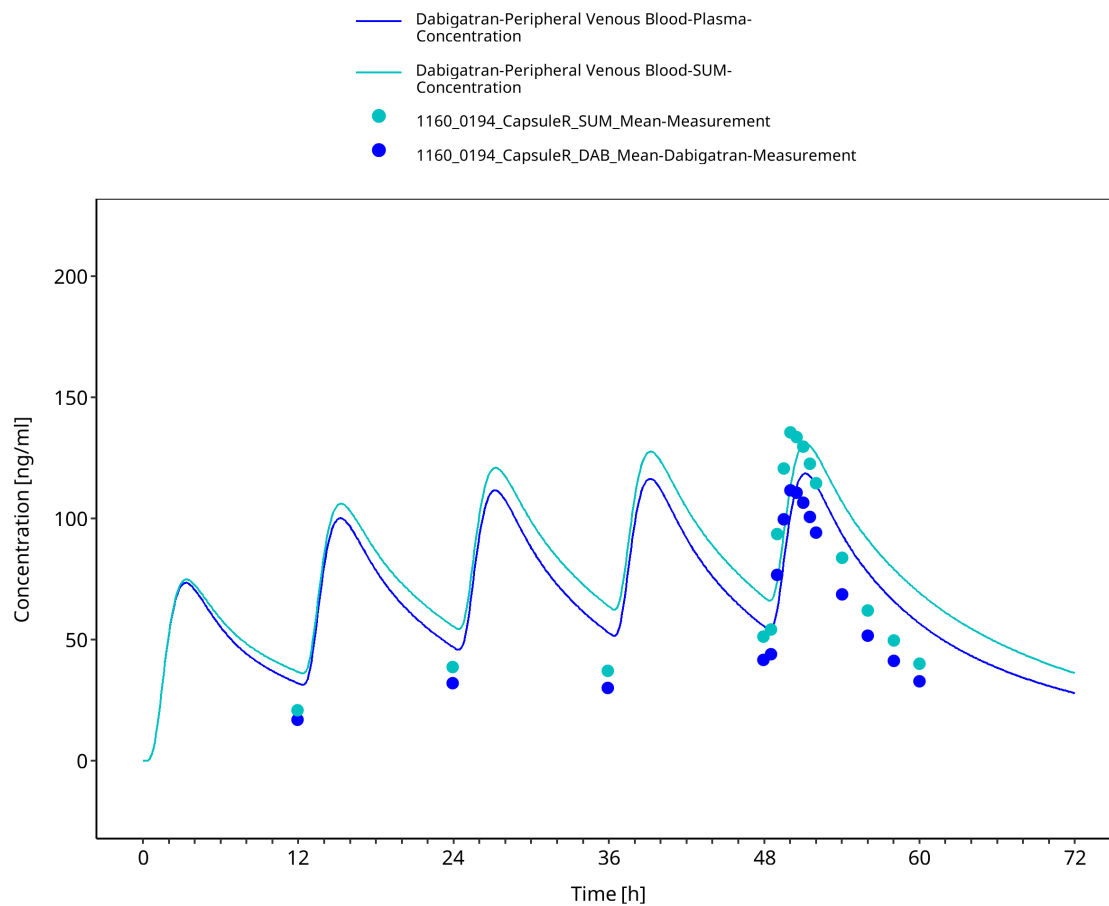


Figure 3-7: 1160-0194 - Dabigatran 150 mg Capsule - lin

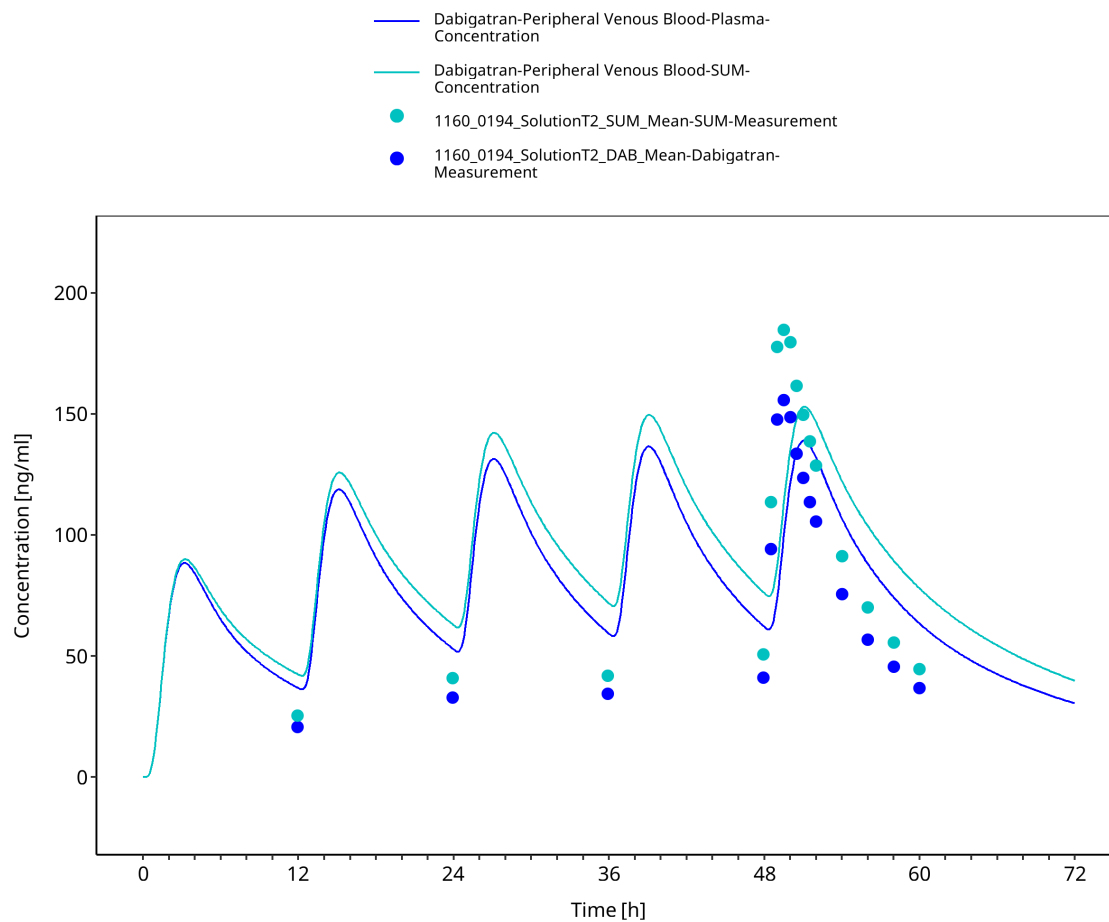


Figure 3-8: 1160-0194 - Dabigatran 150 mg Solution - lin

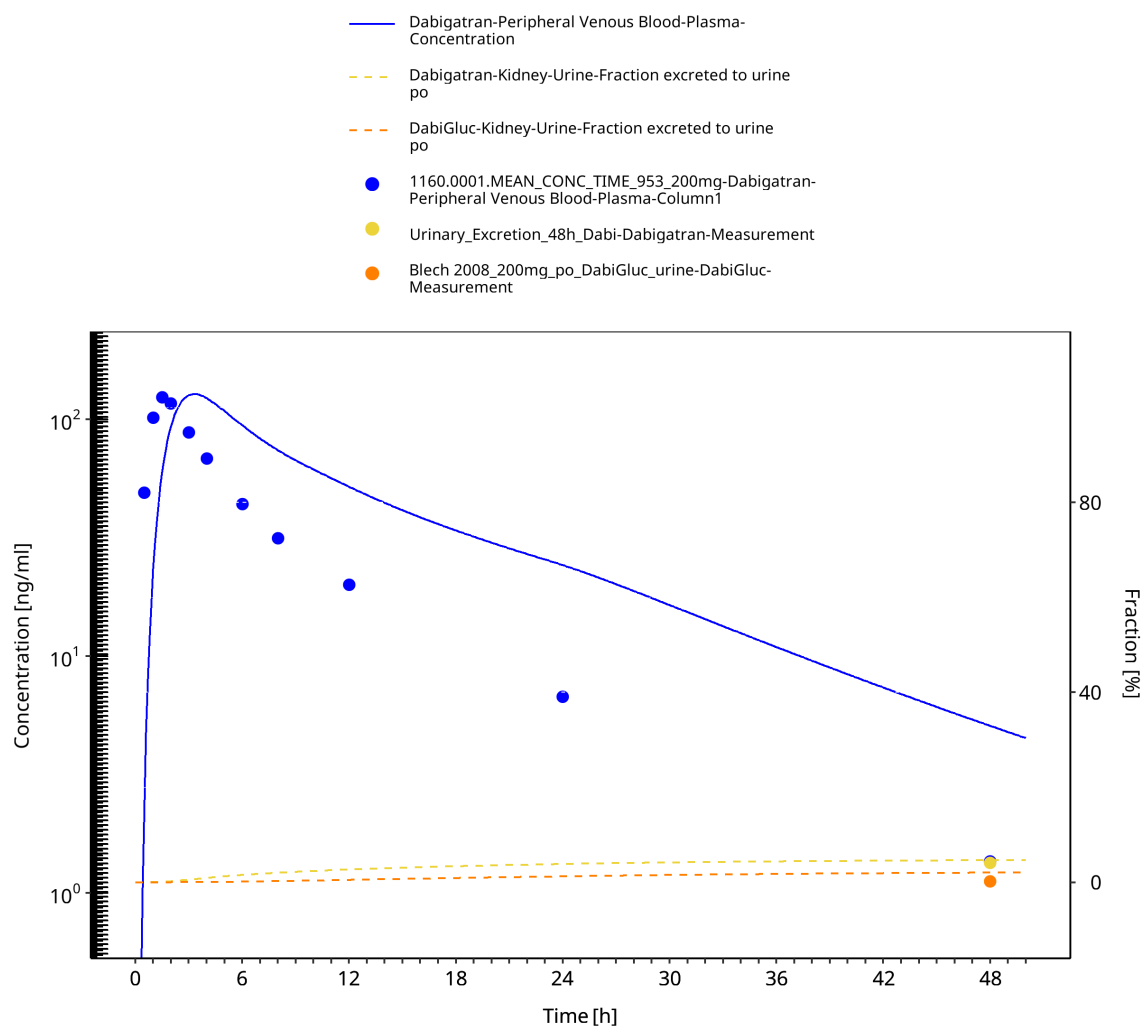


Figure 3-9: Blech 2008 - Dabigatran 200 mg Solution - log

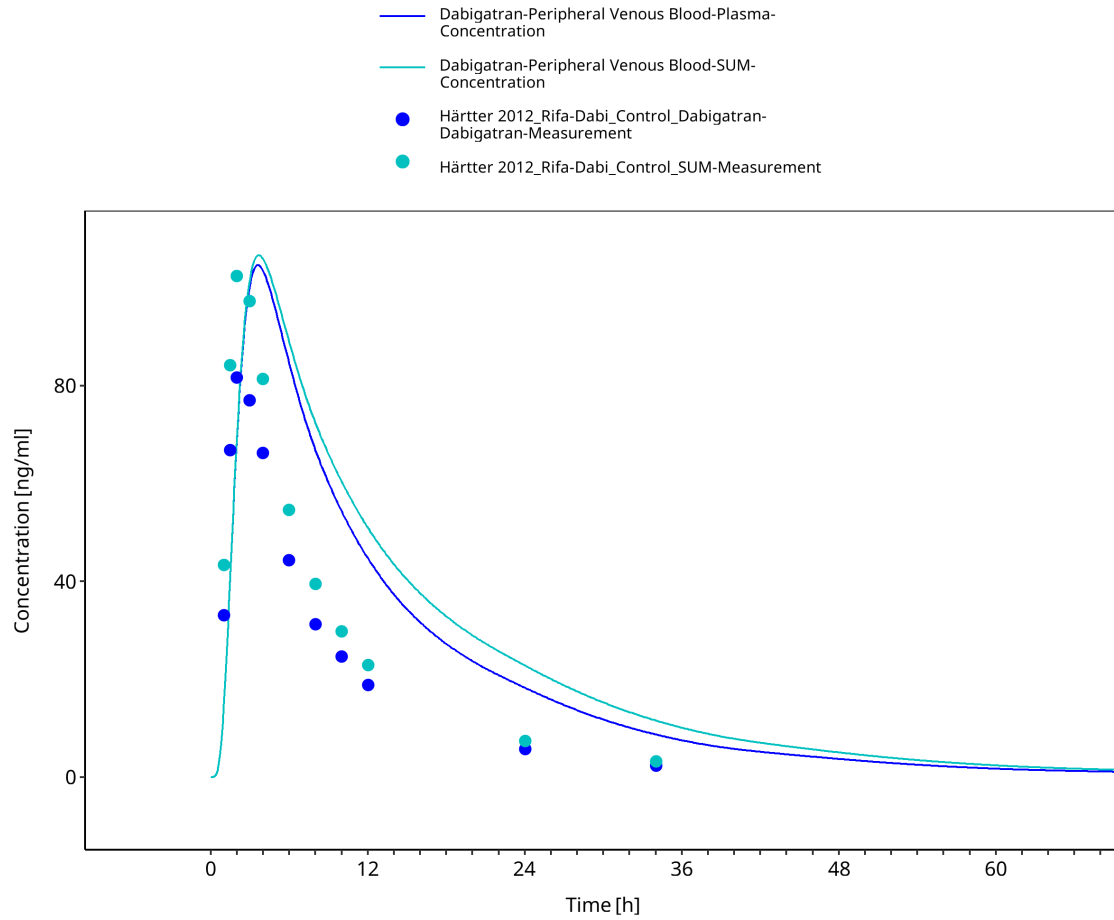


Figure 3-10: Härtter 2012 - Dabigatran DDI Rifampicin - lin

3.3.2 Model Verification

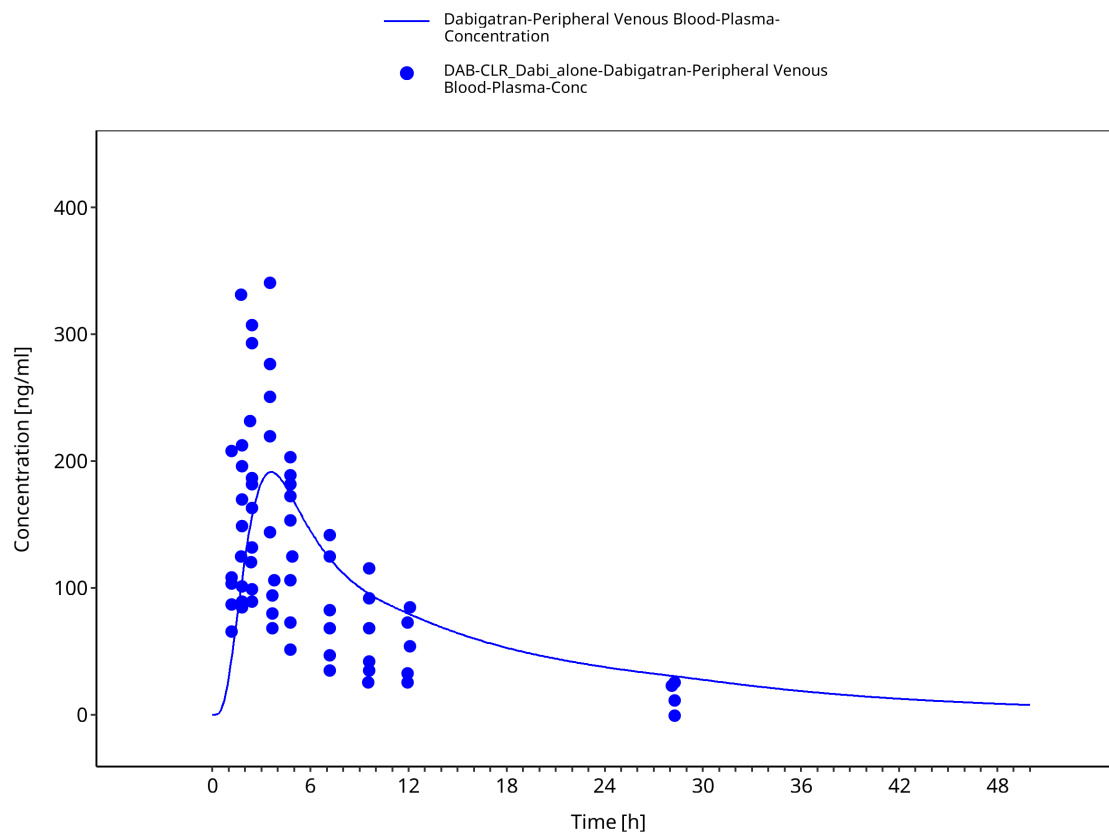


Figure 3-11: Delavenne 2012 - Dabigatran DDI Control - lin

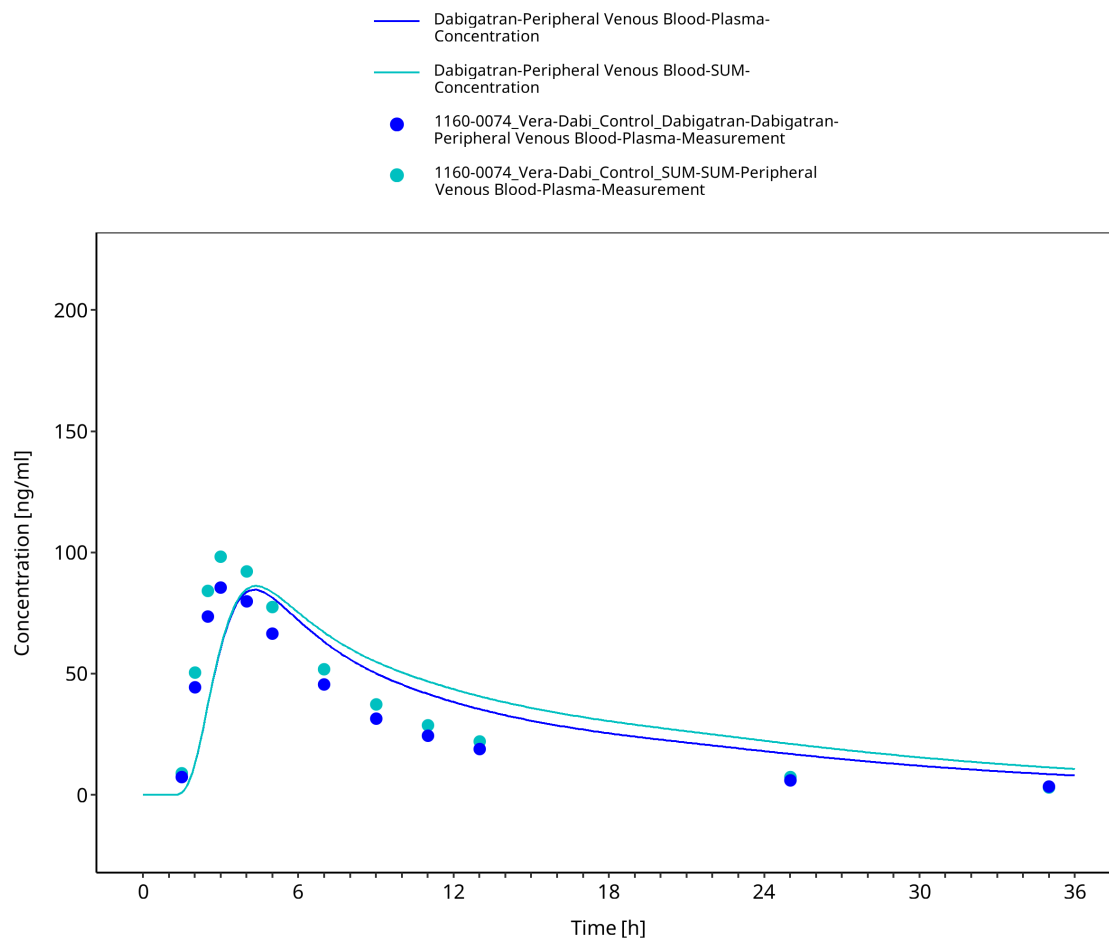


Figure 3-12: 1160-0074 - Dabigatran DDI Control - lin

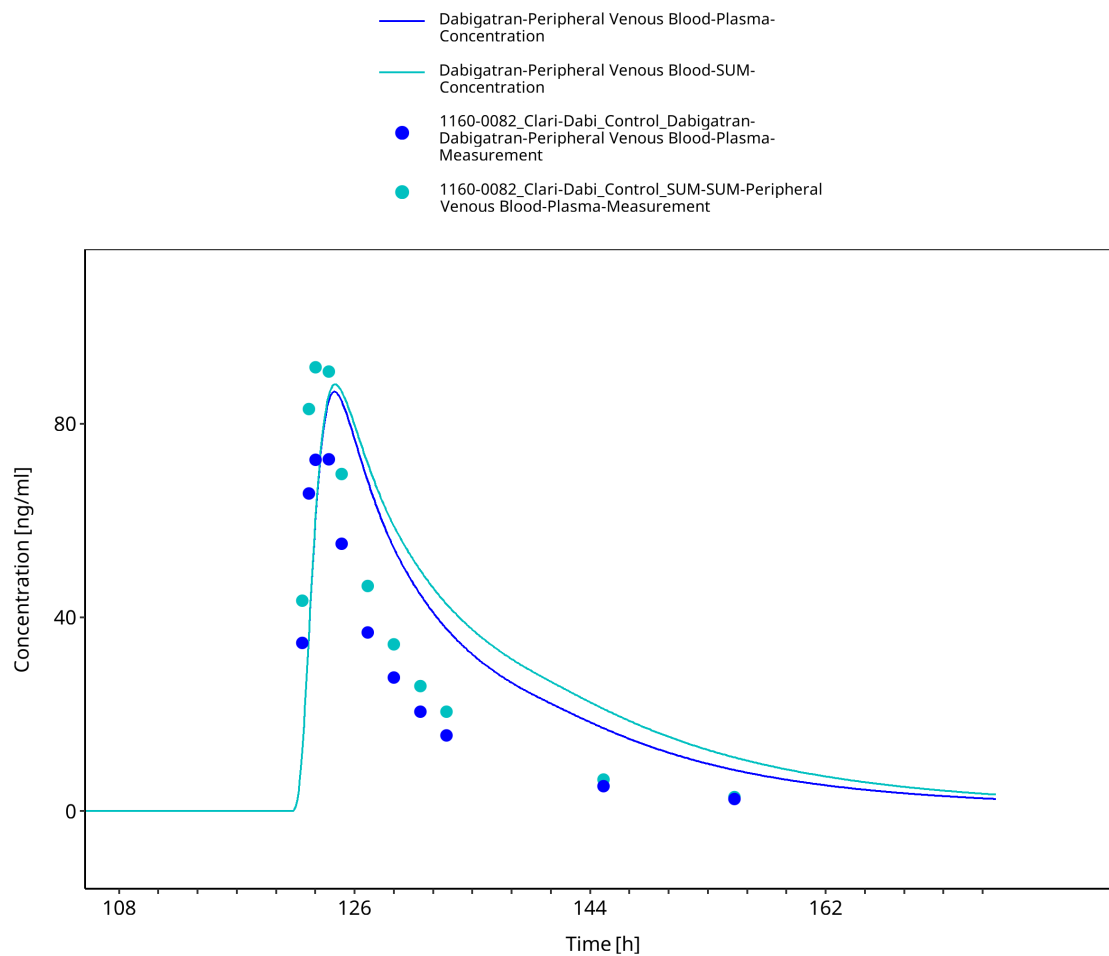


Figure 3-13: 1160-0082 - Dabigatran DDI Control - lin

4 Conclusion

The herein presented PBPK model adequately describes the pharmacokinetics of dabigatran etexilate, dabigatran, and dabigatran glucuronide in healthy adults.

In particular, it integrates key ADME processes including the impact of Pgp on its absorption. Thus, the model is fit for purpose to be applied for the investigation of drug-drug interactions with regard to its Pgp transport.

5 References

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PK-Sim Open Systems Pharmacology Suite PK-Sim® and MoBi® Documentation. (<https://github.com/Open-Systems-Pharmacology/OSPSuite.Documentation>)